



Australian
National
University

Research School
of Earth Sciences



**Remote Sensing of
Water Resources**
EMSC3025/6025

Groundwater — Principles

EMSC3025/6025

[Back to water-course](#)

EMSC3025/6025: Remote Sensing of Water
Resources

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Objectives

By the end of this lecture, you should be able to:

- Understand the **theory of saturated groundwater flow**.
- Derive and apply the **basic groundwater flow equation**.
- Use **flow nets** to estimate hydraulic head distribution and flow paths.
- Identify **boundary and initial conditions** for analytical solutions.
- Solve **simple steady-state flow** problems using analytical methods.

A refresher on some basic math

Understanding Differential Equations in Groundwater

Many students find the quantitative aspects of groundwater hydrology challenging at first glance. 🤔

A differential equation like this immediately suggests advanced mathematical concepts:

$$\frac{\partial}{\partial x} \left(T_x \frac{\partial h}{\partial x} \right) + \frac{\partial}{\partial y} \left(T_y \frac{\partial h}{\partial y} \right) + \frac{\partial}{\partial z} \left(T_z \frac{\partial h}{\partial z} \right) = S \frac{\partial h}{\partial t}$$

Key Point

Hydrogeology provides simplified approaches for dealing with these equations.

Good news: At this introductory level, we don't need to solve these equations—just recognise and understand them.

Recognising Equation Features

You don't need to be a math person for dealing with groundwater equations. Even as toddlers, we could recognise a rhinoceros by its features:

- Four stubby legs
- Two horns on an ugly-looking head

Similarly, differential equations have **recognizable features** that tell us what we're dealing with.

The unknown we're trying to find is usually hidden in derivative terms like:

$$\frac{\partial h}{\partial x} \text{ or } \frac{\partial h}{\partial y} \text{ or } \frac{\partial h}{\partial z}$$

where h is our unknown (hydraulic head).

What Makes an Equation What It Is?

The **unknown variable** determines the equation type:

- h (hydraulic head) → **groundwater flow equation**
- C (concentration) → **mass transport equation**
- T (temperature) → **energy transport equation**

The “stuff” on the right-hand side are functions of time and space variables (t, x, y, z) and various parameters.

Determining Problem Characteristics

Example Analysis:

Step 1: Identify the unknown

- What variable are we solving for?

$$\frac{\partial^2 h}{\partial x^2} = 0$$

Step 2: Count space dimensions

- How many spatial directions does flow occur in?

Step 3: Check for time dependence

- Does the solution change with time?

- Unknown: $h \rightarrow$ groundwater flow
- Dimensions: Only $x \rightarrow$ 1D problem
- Time: No $\frac{\partial h}{\partial t} \rightarrow$ steady-state
- Parameters: None $\rightarrow$ simple case

Problem Dimensionality

Dimensionality describes how many directions the unknown (hydraulic head) changes.

- **1D:** Head changes in one direction only
- **2D:** Head changes in two directions
- **3D:** Head changes in three directions

Important: This refers to how many spatial variables appear in the equation, not the physical geometry.

Example

A 3D aquifer between two rivers might still have 1D flow if water only moves in the x-direction.

Steady-State vs. Transient

Steady-State:

- No time term: $\frac{\partial h}{\partial t} = 0$
- Hydraulic head doesn't change with time
- Easier to solve

Transient:

- Time term present: $\frac{\partial h}{\partial t} \neq 0$
- Hydraulic head changes with time
- More complex solutions

Practice Question

Look at your flow equation. Is there a $\frac{\partial h}{\partial t}$ term?

- **Yes** → Transient problem
- **No** → Steady-state problem

Parameter Dependencies

Sometimes equations contain **parameters** like K, T, or S.

- If parameters are **constant everywhere** → hydraulic head distribution is independent of parameter values
- If parameters **vary in space** → hydraulic head depends on parameter values

Practical impact: You can solve for the pattern of flow, then scale it based on actual parameter values.

Differential Equations of Groundwater Flow

What sets hydrogeology apart is its emphasis on **mathematical treatment** of physical processes.

The goal: determine how **hydraulic head** varies throughout a domain.

Three Essential Components:

1. **Governing equation** – describes groundwater flow
2. **Domain** – geometry and layering
3. **Boundary conditions** – head or flux at boundaries

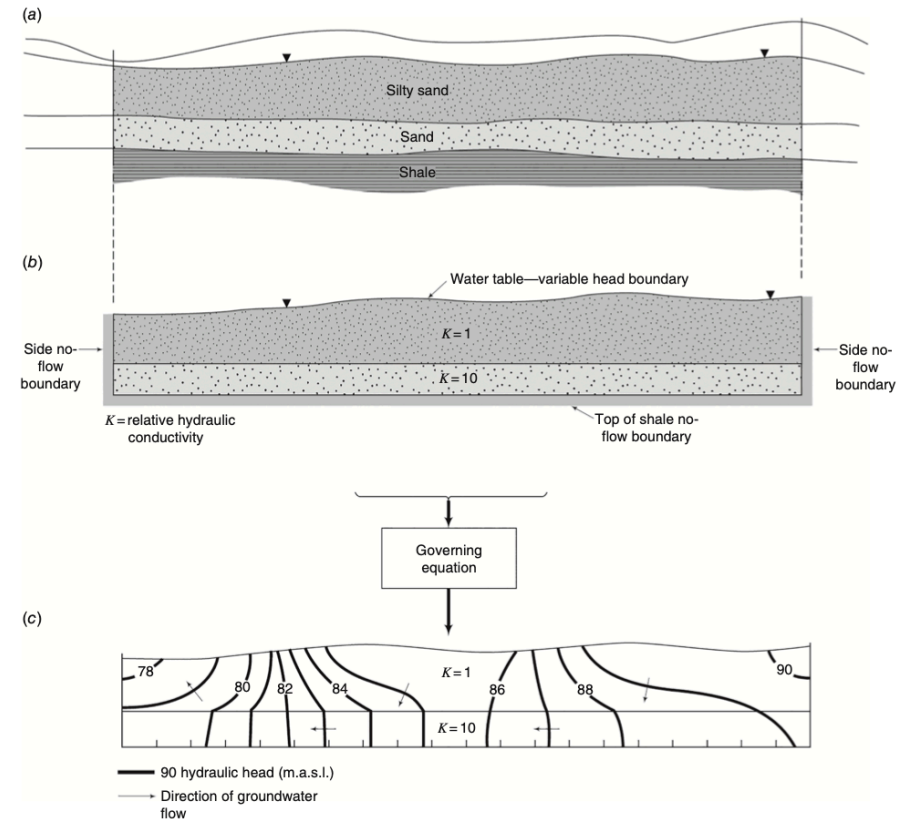
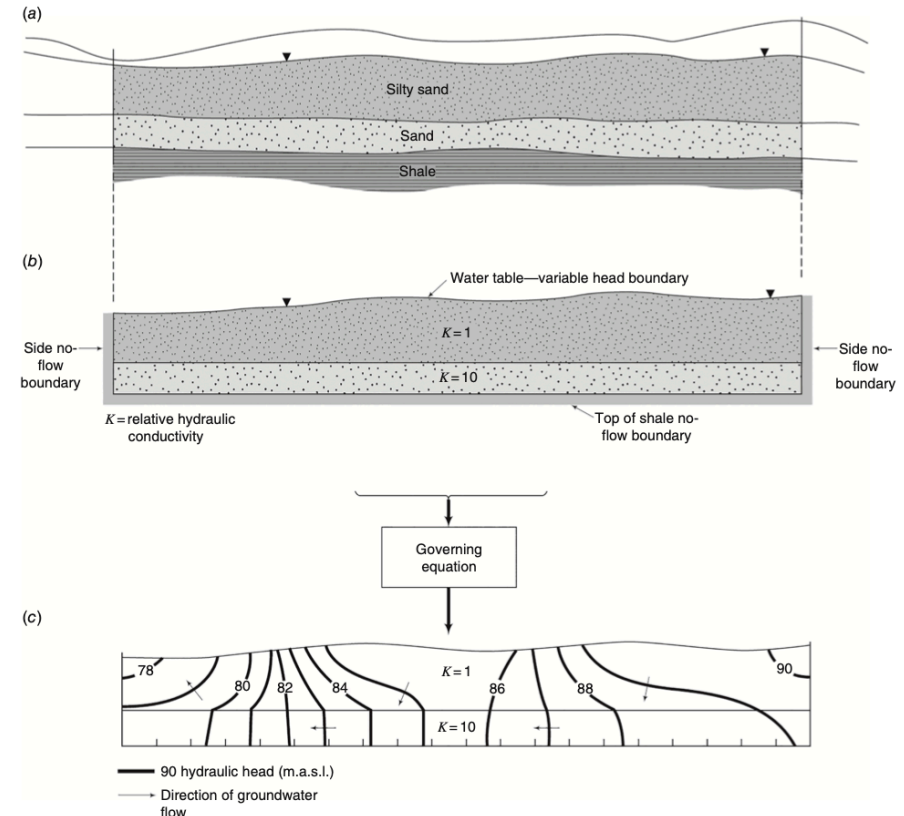


Figure 1: From physical system to solution: geometry → equation → head → flow pattern

What Do We Get From Solving the Equation?

Once we solve for the hydraulic head distribution, we can:

- **Draw equipotentials** (contours of equal head)
- **Deduce flow paths** (direction of groundwater movement)
- **Calculate flow rates** between different points
- **Predict impacts** of pumping or recharge



Flow Domain vs. Flow Equation Dimensionality

Important Distinction:

The dimensionality of the **flow domain** $\neq$ dimensionality of the **flow equation**

- A **1D equation** can describe flow in a **3D domain** if flow occurs in one direction only
- The **actual direction of water movement** determines equation dimensionality

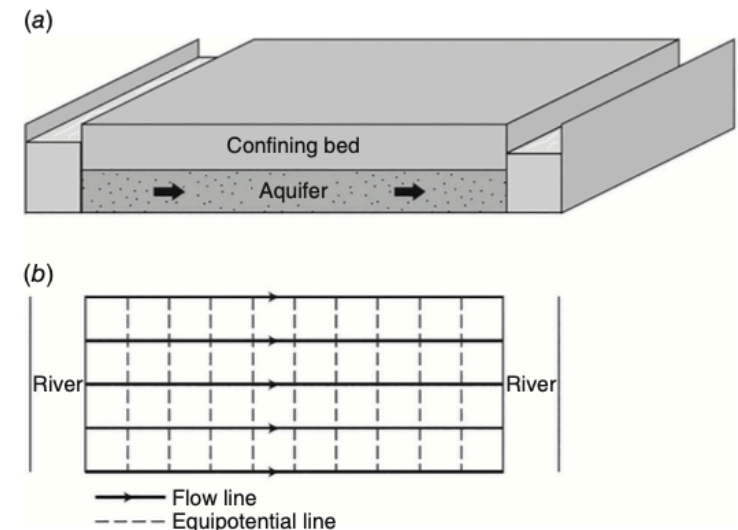


Figure 2: Flow is 1D even in a 3D system

Example: 1D Flow in 3D Domain

Case Study: Aquifer Between Two Rivers

- **Physical structure:** 3D confined aquifer
- **Flow direction:** Only in x-direction (river to river)
- **Governing equation:** 1D flow equation
- **Why?** No head variation in y or z directions

$$\frac{\partial}{\partial x} \left(T_x \frac{\partial h}{\partial x} \right) = S \frac{\partial h}{\partial t}$$

Key Insight:

Look at the **flow pattern**, not the **aquifer geometry**, to determine equation dimensionality.

Deriving Groundwater Flow Equations

Starting Point: Conservation of Mass

Groundwater flow equations come from **mass conservation** applied to a **representative elementary volume (REV)**.

Think of a small cube of aquifer material — water flows in and out through all six faces.

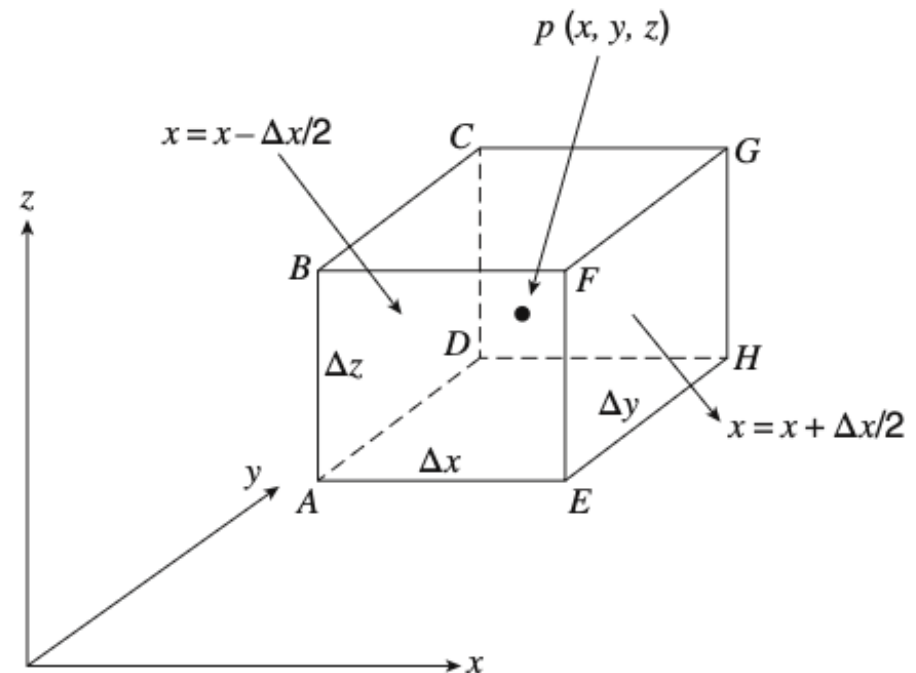


Figure 3: Control volume to derive the equation

The Conservation Principle

$$\text{mass inflow} - \text{mass outflow} = \text{change in storage over time}$$

What we account for:

- Flow through all six faces of the REV
- Water density (ρ_w)
- Porosity (n)
- Velocity components in x, y, z directions

Physical meaning:

- If more water flows in than out $\rightarrow$ head rises
- If more water flows out than in $\rightarrow$ head drops
- If inflow = outflow $\rightarrow$ steady-state

Mathematical Representation of Conservation

$$M_{x1} - M_{x2} + M_{y1} - M_{y2} + M_{z1} - M_{z2} = \frac{\partial}{\partial t} (\rho_w n \Delta x \Delta y \Delta z)$$

Breaking down the equation:

- M_{x1}, M_{x2} = mass flow in x-direction (in and out)
- M_{y1}, M_{y2} = mass flow in y-direction (in and out)
- M_{z1}, M_{z2} = mass flow in z-direction (in and out)
- Right side = rate of mass accumulation in the REV

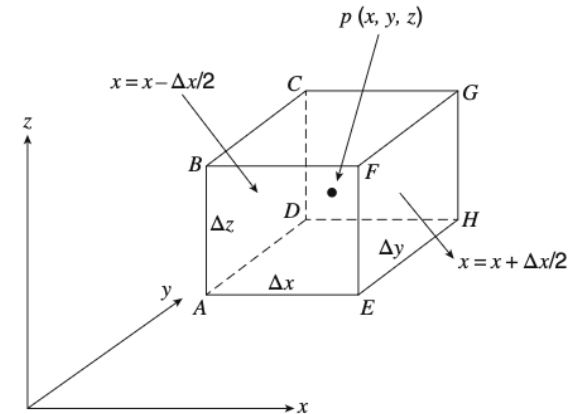


Figure 4: Control volume

Translation: Net mass inflow = rate of mass

Mass Flux Derivation: Step-by-Step

X-Direction Flow

Consider flow entering and leaving the REV in the x-direction:

Inflow at left face:

$$M_{x1} = \left[\rho_w q_x - \frac{1}{2} \frac{\partial(\rho_w q_x)}{\partial x} \Delta x \right] \Delta y \Delta z$$

Outflow at right face:

$$M_{x2} = \left[\rho_w q_x + \frac{1}{2} \frac{\partial(\rho_w q_x)}{\partial x} \Delta x \right] \Delta y \Delta z$$

Key Concept:

Flow varies across the REV due to spatial gradients. We use Taylor expansion around the center point.

Net Flow in X-Direction

$$\text{Net x-direction inflow: } M_{x1} - M_{x2} = -\frac{\partial(\rho_w q_x)}{\partial x} \Delta x \Delta y \Delta z$$

Same logic applies to y and

$$M_{y1} - M_{y2} = -\frac{\partial(\rho_w q_y)}{\partial y} \Delta x \Delta y \Delta z$$

and to z direction:

$$M_{z1} - M_{z2} = -\frac{\partial(\rho_w q_z)}{\partial z} \Delta x \Delta y \Delta z$$

Combining All Directions: Full Conservation

$$-\left[\frac{\partial(\rho_w q_x)}{\partial x} + \frac{\partial(\rho_w q_y)}{\partial y} + \frac{\partial(\rho_w q_z)}{\partial z} \right] = \frac{\partial(\rho_w n)}{\partial t}$$

This equation combines:

- Net inflow from all three directions (left side)
- Rate of mass accumulation (right side)

Physical meaning:

Divergence of flow = storage change

Next step: Simplify by assuming water density is constant

Assuming Constant Water Density

$$-\left[\frac{\partial q_x}{\partial x} + \frac{\partial q_y}{\partial y} + \frac{\partial q_z}{\partial z} \right] = \frac{1}{\rho_w} \frac{\partial(\rho_w n)}{\partial t}$$

Why this assumption? For most groundwater problems, water density changes are negligible compared to flow effects.

Introducing Specific Storage

The right-hand side can be simplified using **specific storage** (S_s):

$$\frac{1}{\rho_w} \frac{\partial(\rho_w n)}{\partial t} = S_s \frac{\partial h}{\partial t}$$

Definition: S_s accounts for water release due to both aquifer compression and water expansion.

Remember:

S_s links changes in storage to changes in hydraulic head.

Conservation in Terms of Hydraulic Head

$$\frac{\partial q_x}{\partial x} + \frac{\partial q_y}{\partial y} + \frac{\partial q_z}{\partial z} = -S_s \frac{\partial h}{\partial t}$$

We're almost there! Next step: substitute Darcy's law to express everything in terms of hydraulic head.

Applying Darcy's Law

Darcy's Law in 3D:

- $q_x = -K_x \frac{\partial h}{\partial x}$
- $q_y = -K_y \frac{\partial h}{\partial y}$
- $q_z = -K_z \frac{\partial h}{\partial z}$

Key insight:

Darcy's law relates flow to head gradients using hydraulic conductivity.

Substituting into our conservation equation...

The Final Groundwater Flow Equation

$$\frac{\partial}{\partial x} \left(K_x \frac{\partial h}{\partial x} \right) + \frac{\partial}{\partial y} \left(K_y \frac{\partial h}{\partial y} \right) + \frac{\partial}{\partial z} \left(K_z \frac{\partial h}{\partial z} \right) = S_s \frac{\partial h}{\partial t}$$

This is the fundamental equation for flow in saturated porous media!

What it describes:

- How hydraulic head (h) varies in space and time
- 3D flow in heterogeneous, anisotropic media

Parameters needed:

- K_x, K_y, K_z = hydraulic conductivity
- S_s = specific storage
- Boundary and initial conditions

Special Case 1: Steady-State Flow

When $\frac{\partial h}{\partial t} = 0$ (no change in head with time):

$$\frac{\partial}{\partial x} \left(K_x \frac{\partial h}{\partial x} \right) + \frac{\partial}{\partial y} \left(K_y \frac{\partial h}{\partial y} \right) + \frac{\partial}{\partial z} \left(K_z \frac{\partial h}{\partial z} \right) = 0$$

Physical meaning:

- System has reached equilibrium
- Inflow = outflow everywhere
- Head pattern is time-invariant
- Much easier to solve than transient problems

Example situations:

- Long-term pumping
- Seasonal averages
- Regional flow systems

Special Case 2: Isotropic, Homogeneous Media

When $K_x = K_y = K_z = K$ (constant) and steady-state:

$$\frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial y^2} + \frac{\partial^2 h}{\partial z^2} = 0$$

This is Laplace's equation!

- Classic partial differential equation
- Many analytical solutions available
- Foundation for flow net analysis
- Widely used in mathematical physics

Simplification:

When properties are uniform, the equation becomes much simpler to solve.

Working with Transmissivity

For **confined aquifers**, we often use **transmissivity**:

- $T_x = K_x \times b$ (transmissivity in x-direction)
- $S = S_s \times b$ (storativity)
- b = aquifer thickness

$$\frac{\partial}{\partial x} \left(T_x \frac{\partial h}{\partial x} \right) + \frac{\partial}{\partial y} \left(T_y \frac{\partial h}{\partial y} \right) = S \frac{\partial h}{\partial t}$$

Why transmissivity?

For confined aquifers, flow is integrated over the thickness, making T more practical.

Including Sources and Sinks

Real systems have **sources and sinks**:

- Wells (pumping or injection)
- Recharge from precipitation
- Leakage to/from other layers

Sign convention:

- $Q > 0$ = source (injection)
- $Q < 0$ = sink (pumping)

Adding source term $Q(x, y, z, t)$:

$$\frac{\partial}{\partial x} \left(K_x \frac{\partial h}{\partial x} \right) + \frac{\partial}{\partial y} \left(K_y \frac{\partial h}{\partial y} \right) + \frac{\partial}{\partial z} \left(K_z \frac{\partial h}{\partial z} \right) + Q(x, y, z, t) = S_s \frac{\partial h}{\partial t}$$

Special Case 3: Unconfined Aquifers

Dupuit's Assumption

For **unconfined aquifers**, we make a simplifying assumption:

Dupuit approximation: Vertical flow is negligible compared to horizontal flow.

- Streamlines are horizontal
- Hydraulic gradient equals water table slope
- Valid when horizontal distances $\gg$ aquifer thickness

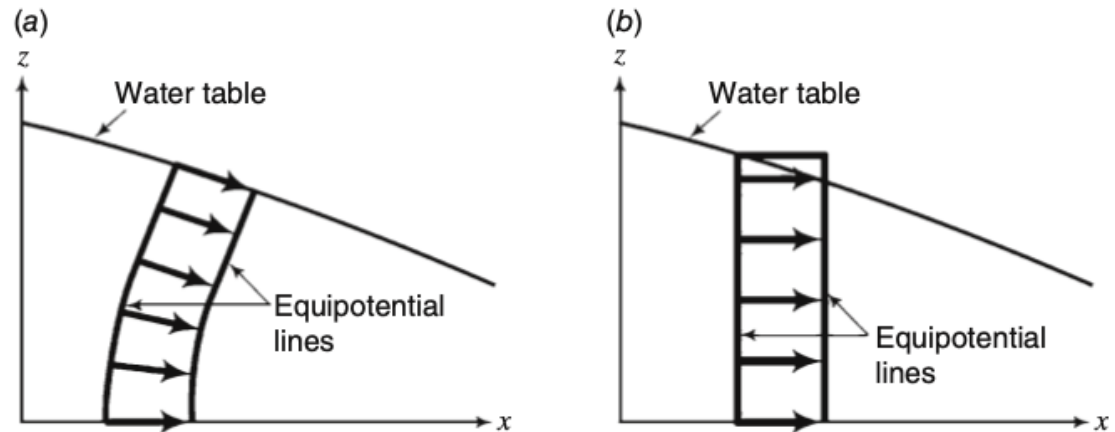


Figure 5: a) actual flow, b) Dupuit approximation

Boussinesq's Equation

Under Dupuit approximation:

$$\frac{\partial}{\partial x} \left(K(x, y) h \frac{\partial h}{\partial x} \right) + \frac{\partial}{\partial y} \left(K(x, y) h \frac{\partial h}{\partial y} \right) = S_y \frac{\partial h}{\partial t}$$

Key features:

- h appears both inside and outside derivatives
- S_y = specific yield (not specific storage)
- Commonly used for water table aquifers

Note:

This is **Boussinesq's equation** - the standard for unconfined flow analysis.

Solving the Groundwater Flow Equation

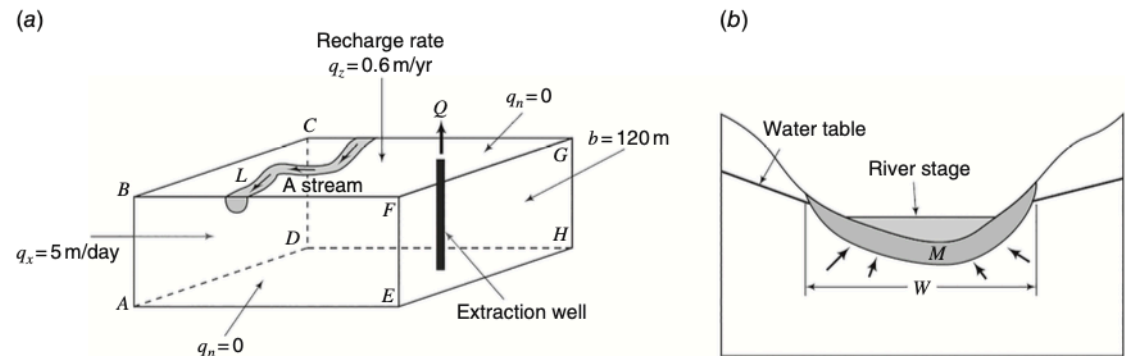
The Missing Piece: Boundary Conditions

The flow equation alone is not enough!

To get a unique solution, we must specify **boundary conditions** - they define how our domain interacts with the surroundings.

Three main types:

1. **Dirichlet** - specify head values
2. **Neumann** - specify flow rates
3. **Cauchy** - relate head and flow



Examples of different boundary condition types

Why Boundary Conditions Matter

Without boundary conditions:

- Infinite number of possible solutions
- No way to determine actual head values
- Cannot predict real system behavior

With proper boundary conditions:

- Unique solution exists
- Realistic head distribution
- Meaningful flow predictions

Key applications:

- Rivers and lakes (known water levels)
- Recharge zones (known flux)
- Wells (extraction rates)
- No-flow boundaries (impermeable layers)

“Specified Head” Boundaries

In math we call this Dirichlet Boundary Conditions

What it specifies: Hydraulic head values directly

$$h(x, y, z)|_{\text{At a boundary}} = h_1(x, y, z, t)$$

Common examples:

- Large lakes or rivers
- Ocean boundaries
- Artificial ponds or reservoirs

When to use:

When you know the water level at a boundary and it's much larger than your study area.

For all of these you can calculate the pressure.

Type 2: Neumann Boundary Conditions

“Specified Flux” Boundaries

What it specifies: Water flux (flow rate) across the boundary

$$q_n \big|_{\text{At a boundary}} = q_n(x, y, z, t)$$

Common examples:

- Precipitation recharge ($q_n > 0$)
- Pumping wells ($q_n < 0$)
- Impermeable boundaries ($q_n = 0$)

Remember $\nabla h \cdot n = q_n$

Special case:

$q_n = 0$ (no-flow) is very common for impermeable rock or clay layers.

Understanding Flux Units

Flux can be expressed two ways:

1. Darcy velocity q_n [L/T]

- Flow per unit area
- [This is the source term we use the flow equation](#)

2. Volumetric flow rate Q_n [L^3/T]

- Total volume per time
- Used for wells and boundaries

Relationship:

$$Q_n = q_n \times \text{Area}$$

“Head-Dependent Flux” Boundaries (Cauchy)

Sometimes the amount of flux in or out of the system is depending on the solution itself.

What it specifies: Relationship between flux and head difference

$$q_n = \frac{K_m}{M} [h_m(x, y, z, t) - h(x, y, z, t)]$$

Parameters:

- K_m = hydraulic conductivity of boundary layer
- M = thickness of boundary layer
- h_m = external head (e.g., river stage)

Key feature:

Flux depends on head difference — more realistic for many situations.

When to Use Cauchy Boundaries

Ideal for river-aquifer interactions:

- River stage varies with time (h_m changes)
- Riverbed acts as semi-permeable layer
- Flow direction can reverse (gaining vs. losing stream)

Advantages:

- More physically realistic
- Automatically handles flow reversals
- Accounts for resistance of boundary layer

Example:

During floods, $h_m > h$ so water flows from river to aquifer. During droughts, $h > h_m$ so aquifer feeds the river.

Choosing the Right Boundary Condition

Decision Matrix:

Dirichlet

- Large water bodies
- Known, stable water levels
- Simple to implement

Neumann

- Known recharge rates
- Impermeable boundaries
- Well pumping/injection

Cauchy

- River-aquifer exchange
- Variable external conditions
- Semi-permeable boundaries

The choice depends on: physical setting, available data, and modeling objectives.

Initial Conditions: Starting Point for Transient Problems

For **transient (time-dependent) problems**, we need to specify where we start:

$$h(x, y, z, 0) = h_0(x, y, z)$$

What this means: At time zero, the hydraulic head at every point in the domain must be known.

Important:

Steady-state problems don't need initial conditions — they find the equilibrium directly.

Why Initial Conditions Matter

Examples of initial conditions:

- Aquifer at natural equilibrium before pumping
- Known water table elevation from well measurements
- Results from previous model run

Physical meaning: The system's “starting configuration” before any changes occur.

Practical tip:

Often use steady-state solution as initial condition for transient problems.

Flow-net Analysis

Constructing the streamlines or the flowlines

Visualising Groundwater Flow

From Equations to Pictures

We can **visualise the results** using flow nets:

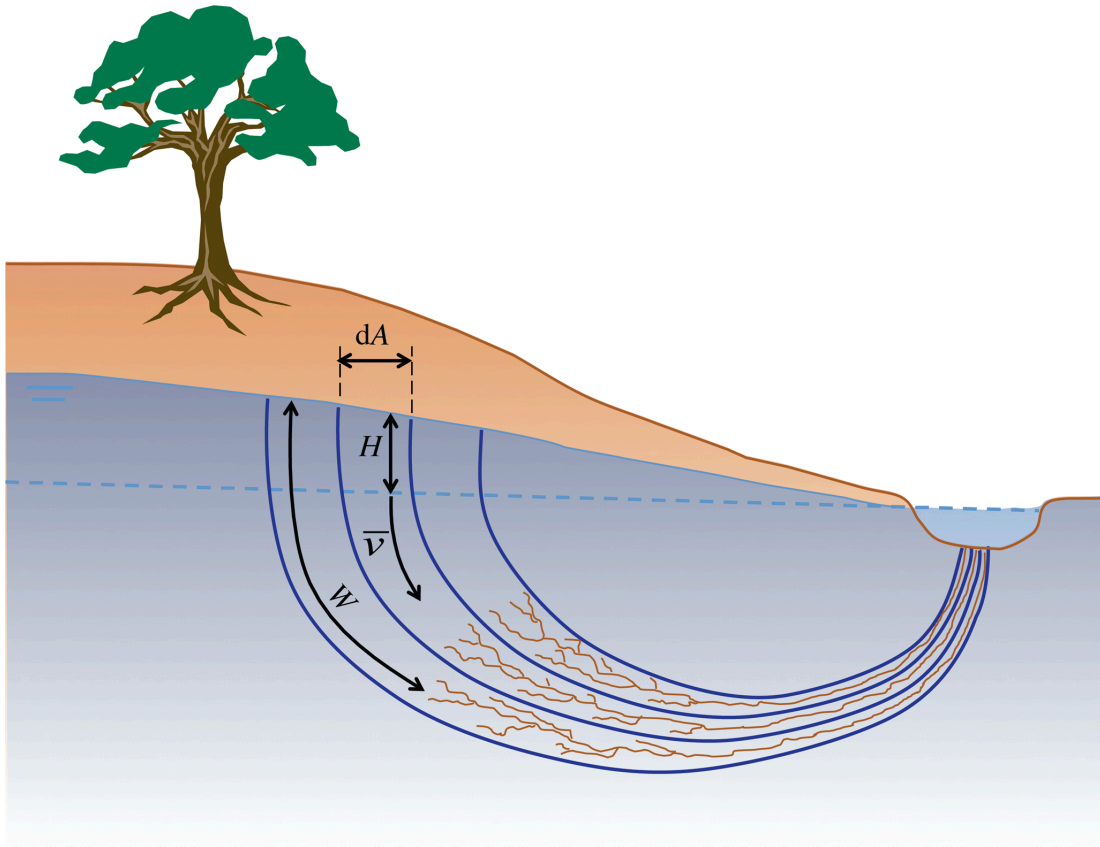
Two key components:

- **Streamlines** (flow paths): show direction of water movement
- **Equipotential lines**: contours of equal hydraulic head

Key property: In isotropic, homogeneous media, streamlines and equipotentials intersect at **right angles**.

Remember:

Water flows perpendicular to equipotential lines, from high head to low head.



Flowlines are our way of visualising the flow, without solving the flow equation.

The Mathematical Foundation

For **2D steady-state flow** in isotropic, homogeneous media:

$$\frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial y^2} = 0$$

This is **Laplace's equation** - it has special mathematical properties that make flow nets possible.

Why 2D?

Many practical problems can be simplified to 2D, making flow nets a powerful tool.

Rules for Constructing Flow Nets

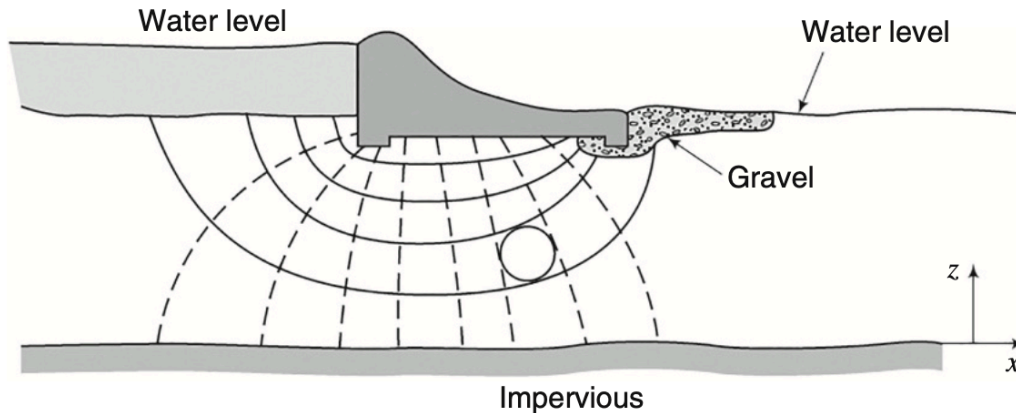
Geometric rules:

1. **Orthogonality:** Equipotentials $\perp$ streamlines
2. **Curvilinear squares:** Lines form approximate squares
3. **Equal spacing:** Same head drop between equipotentials
4. **Conservation:** Same flow between adjacent streamlines

Boundary conditions:

- **No-flow boundaries** $\rightarrow$ parallel to streamlines
- **Constant-head boundaries** $\rightarrow$ parallel to equipotentials
- **Symmetry lines** $\rightarrow$ can be treated as no-flow

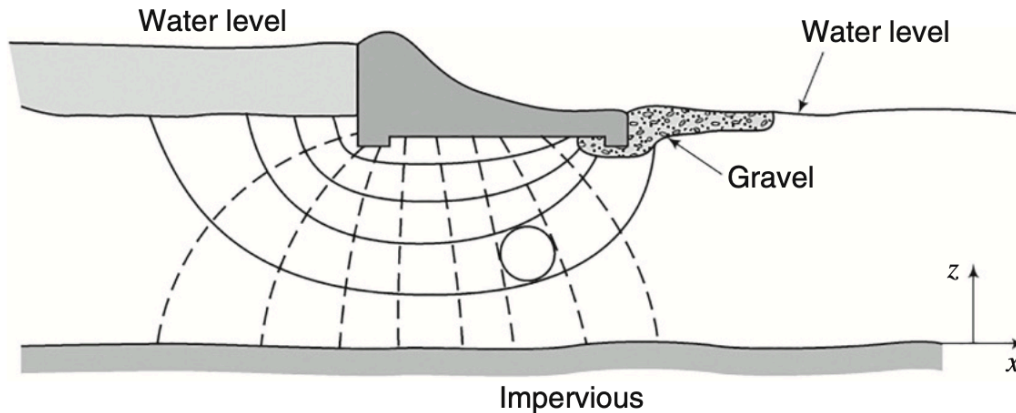
The famous dam seepage problem



Practical tip:

Figure 6: Seepage through permeable foundation

The famous dam seepage problem



Practical tip:

Start with boundary conditions from page before, then sketch equipotentials and streamlines iteratively until squares form.

Figure 6: Seepage through permeable foundation

The famous dam seepage problem

Classic problem: seepage under a dam

Boundary conditions:

- **Left side:** Constant head (upstream pool level)
- **Right side:** Constant head (downstream pool level)
- **Bottom:** No-flow boundary (impermeable bedrock)
- **Top:** Free surface or no-flow

Result: Curved flow paths that form approximate squares

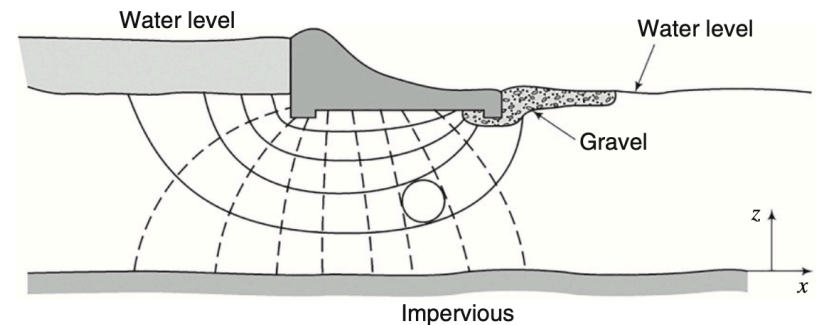


Figure 7: Seepage through permeable foundation

Reading the Flow Net

What the flow net tells us:

- **Flow direction:** Follow the streamlines
- **Head distribution:** Read from equipotential values
- **Flow concentration:** Closer streamlines = higher velocity
- **Seepage quantity:** Can be calculated from the net

Engineering applications:

- Assess dam safety (uplift pressure)
- Design drainage systems
- Estimate seepage losses

Key insight:

Maximum head gradient (steepest flow) occurs where equipotentials are closest together.

Quantifying flow

Let's consider the 1 dimensional example

For a single flow channel between two streamlines:

consider the 1d Darcy equation $Q = Aq = -AK \frac{\partial h}{\partial x}$

$$\Delta Q = T \Delta h \frac{\Delta W}{\Delta L}$$

Where:

- ΔQ = flow through one channel
- T = transmissivity
- Δh = head drop between equipotentials
- $\Delta W / \Delta L$ = width/length ratio (≈ 1 for squares)

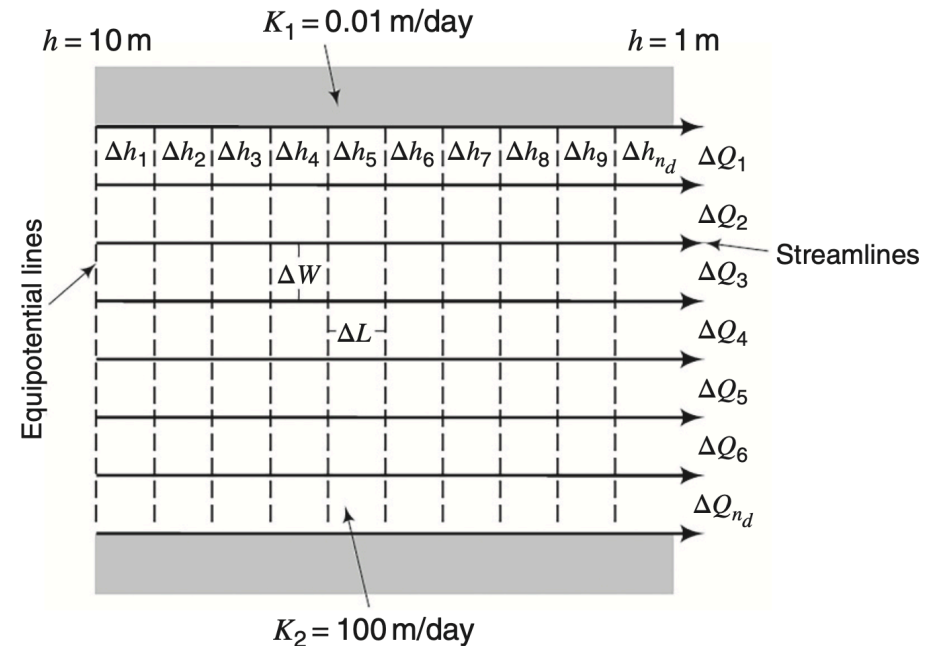


Figure 8: Streamlines and equipotentials under homogenous and isotropic conditions, flow is due to the constant heads applied at the left and right sides of the domain

Another way of looking at it

Let's say there are a total number of n_f flow channels and n_d equipotential drops.

$$\Delta Q = \frac{n_f}{n_d} T \Delta h$$

then the flow rate is equal to $n_f \Delta Q$ and pressure drop is equal to $n_d \Delta h$.

Where:

- ΔQ = flow through one channel
- T = transmissivity
- Δh = head drop between equipotentials
- $\Delta W / \Delta L$ = width/length ratio (≈ 1 for squares)

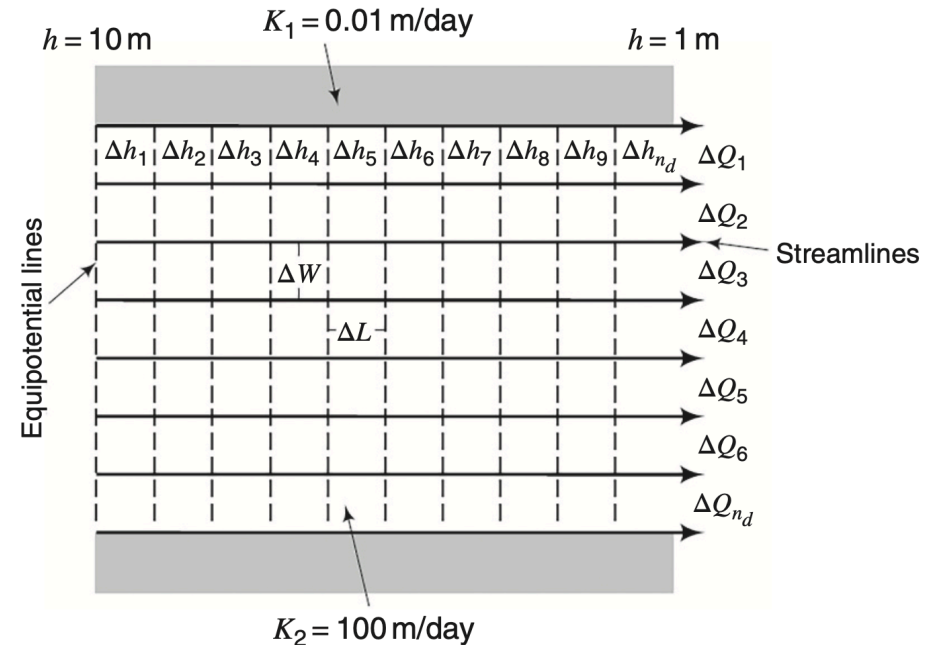


Figure 9: Streamlines and equipotentials under homogenous and isotropic conditions, flow is due to the constant heads applied at the left and right sides of the domain.

Total Flow Calculation

For the entire flow system:

$$Q_{total} = \frac{n_f}{n_d} T \Delta H$$

Where:

- n_f = number of flow channels (between streamlines)
- n_d = number of equipotential drops
- ΔH = total head difference across system

Practical steps:

1. Count flow channels
2. Count equipotential intervals
3. Measure total head drop
4. Apply formula

What we learned

We explored the theoretical foundations of groundwater flow, covering:

- **Mathematical Framework:** recognizing differential equations and their characteristics (dimensionality, steady-state vs transient)
- **Equation Derivation:** conservation of mass in a representative elementary volume leading to the groundwater flow equation
- **Darcy's Law Integration:** combining flow relationships with conservation principles
- **Specialized Cases:** steady-state flow, isotropic media (Laplace's equation), and unconfined aquifers (Boussinesq equation)
- **Boundary Conditions:** Dirichlet (specified head), Neumann (specified flux), and Cauchy (head-dependent flux) boundaries
- **Initial Conditions:** starting points for transient problems and their physical significance
- **Flow Net Analysis:** visual representation of flow patterns using streamlines and equipotentials
- **Practical Applications:** dam seepage, flow quantification, and engineering design using flow nets